11. Sequences and Series

11.10. Taylor and Maclaurin Series

11.10 Learning Objectives
- Find the Taylor Series Expansion of a function
- Find the Maclaurin Series Expansion of a function
- Find the function represented by the Taylor or Maclaurin Series
- Use known Maclaurin Series Expansion of functions to describe complicated functions

11.10.1. Motivation: Linearization of a function

Recall the linearization of a function

Definition 11.10.1 Linearization of a function

For a (continuous near a) function f, the linearization of f centered at a is the tangent line to the function at the point $(a, f(a))$. Precisely, in *point slope* form ($y = m(x - x_1) + y_1$)

$$L(x) = f'(a)(x - a) + f(a)$$

todo: graph of linearization

- The idea behind linearization is to come up with a simple function (a line L) that matches f in function value and first derivative.
- What's better than matching the function in function value and first derivative at $x = a$?
  - Matching the function in value, first derivative, *and* second derivative at $x = a$.
- What's better than matching the function in value, first derivative, and second derivative at $x = a$?
  - Matching the function in value, first derivative, second derivative, *and* third derivative at $x = a$
  - ...
- Then we build up from there! Going one more degree at a time, matching one extra derivative at $x = a$ each term!

11.10.2. Taylor Series and Maclaurin Series

We have been able to write the power series of some functions using algebraic manipulations to line it up with $\frac{1}{1-u}$ type things. However, we may want to represent other functions with power series.

Suppose that f can be represented with a power series. Then,

$$f(x) = \sum_{n=0}^{\infty} c_n(x-a)^n = c_0 + c_1(x-a) + c_2(x-a)^2 + c_3(x-a)^3 + c_4(x-a)^4 + \ldots$$

We already know that when we evaluate this at $x = a$ we get

$$f(a) = c_0$$

Now, taking the element-wise derivative we see that

$$f'(x) = c_1 + 2 \cdot c_2(x-a) + 3 \cdot c_3(x-a)^2 + 4 \cdot c_4(x-a)^3 + \ldots$$
$$\text{so} \quad f'(a) = c_1$$

Taking another derivative we see

$$f''(x) = 2 \cdot 1 \cdot c_2 + 3 \cdot 2 \cdot c_3(x-a) + 4 \cdot 3 \cdot c_4(x-a)^2 + \ldots$$
$$\text{so} \quad f'(a) = 2c_1$$
$$\text{or} \quad c_1 = \frac{f'(a)}{2}$$

Taking another derivative we see that

$$f'''(x) = 3 \cdot 2 \cdot 1 \cdot c_3 + 4 \cdot 3 \cdot 2 \cdot c_4(x-a) + \ldots$$
$$\text{so} \quad f'''(a) = 3!c_3$$
$$\text{or} \quad c_3 = \frac{f'''(a)}{3!}$$

Doing one more to really see the pattern,

$$f^{(4)}(x) = 4 \cdot 3 \cdot 2 \cdot 1 \cdot c_4 + 5 \cdot 4 \cdot 3 \cdot 2 \cdot c_5(x-a) + \ldots$$
$$\text{so} \quad f^{(4)}(a) = 4!c_4$$
$$\text{or} \quad c_4 = \frac{f^{(4)}(a)}{4!}$$

In general, the nth coefficient must be

$$c_n = \frac{f^{(n)}(a)}{n!}$$

Theorem 11.10.1

If a function f has a power series representation, then

$$f(x) = \sum_{n=0}^{\infty} \frac{f^{(n)}(a)}{n!}(x-a)^n$$

Definition 11.10.2 **Taylor Series**

The series

$$f(x) = \sum_{n=0}^{\infty} \frac{f^{(n)}(a)}{n!}(x-a)^n$$

is called the **Taylor Series of f centered at a**.

This series is named after Brook Taylor, who wrote a paper using Taylor Series in 1715, but the importance of it wasn't noticed until 1772 when Lagrange stumbled across it.

Definition 11.10.3

If $a = 0$, then we say the series

$$f(x) = \sum_{n=0}^{\infty} \frac{f^{(n)}(a)}{n!}x^n$$

is the **Maclaurin Series of f**.

This series is named after Colin Maclaurin, who did a lot of important mathematics using a specialized form of Taylor's series (at $a = 0$). I had fun reading his Wikipedia page, and learning that we have already seen another result of Maclaurin's! And that is Simpson's Rule! (behind the scenes Simpson's Rule comes from integrating a truncated Maclaurin series to approximate the integral!) Also, Maclaurin was a child prodigy who held the record for youngest math professor for centuries until Alia Sabur became a university professor when she was only 18 years old!

Example 11.10.1

Find the Maclaurin series, and its radius of convergence, for the function $f(x) = e^x$.

Example 11.10.2

Find the Taylor Series Expansion for $f(x) = \cos(x)$ centered at $x = \pi$, and find its radius of convergence.

11.10.3. Maclaurin Series of Common Functions

Exercise 11.10.1

Derive the above Maclaurin series expansions and intervals of convergence for the above series.

11.10a Section Summary: